Q1 - 2024 (27 Jan Shift 2)

The integral $\int \frac{(x^8-x^2)dx}{(x^{12}+3x^6+1)\tan^{-1}\left(x^3+\frac{1}{x^3}\right)}$ equal to :

(1) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{1/3}+C$

(2) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{1/2}+C$

(3) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)+C$

(4) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{3}+C$

Q2 - 2024 (29 Jan Shift 1)

For x $\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, if $y(x)=\int\frac{\operatorname{cosec}x+\sin x}{\operatorname{cosec}x\sec x+\tan x\sin^2 x}dx$ and $\lim_{x\to\left(\frac{\pi}{2}\right)^-}y(x)=0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

(1) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(2) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(3) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(4) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$

Q3 - 2024 (29 Jan Shift 2)

If

$$\int\frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx=A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$$

where C is the integration constant, then AB is equal to

(1) $4\operatorname{cosec}(2\theta)$

(2) $4\sec\theta$

(3) $2\sec\theta$

(4) $8\operatorname{cosec}(2\theta)$

Answer Key

Q1 (1) **Q2** (3) **Q3** (4)

Q1

$$I=\int \frac{x^8-x^2}{\left(x^{12}+3x^6+1\right)\tan^{-1}\left(x^3+\frac{1}{x^3}\right)}dx$$

Let $\tan^{-1}\left(x^3+\frac{1}{x^3}\right)=t$

$$\Rightarrow \frac{1}{1+\left(x^3+\frac{1}{x^3}\right)^2}\cdot\left(3x^2-\frac{3}{x^4}\right)dx=dt$$

$$\Rightarrow \frac{x^6}{x^{12}+3x^6+1}\cdot\frac{3x^6-3}{x^4}dx=dt$$

$$I=\frac{1}{3}\int\frac{dt}{t}=\frac{1}{3}\ln|t|+C$$

$$I=\frac{1}{3}\ln\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|+C$$

$$I=\ln\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|^{1/3}+C$$

Hence option (1) is correct

Q2

$$y(x)=\int\frac{\left(1+\sin^2 x\right)\cos x}{1+\sin^4 x}dx$$

Put $\sin x=t$

$$=\int\frac{1+t^2}{t^4+1}dt=\frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(t-\frac{1}{t}\right)}{\sqrt{2}}+C$$

$$x=\frac{\pi}{2}, t=1 \quad \therefore C=0$$

$$y\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{-1}{\sqrt{2}}\right)$$

Q3

$$\int \frac{\sin^{\frac{3}{2}} x+\cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx$$

$$I=\int \frac{\sin^{\frac{3}{2}} x+\cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x(\sin x \cos \theta-\cos x \sin \theta)}} dx$$

$$=\int \frac{\sin^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x \cos^2 x \sqrt{\tan x \cos \theta-\sin \theta}} dx+\int \frac{\cos^{\frac{3}{2}} x}{\sin^2 x \cos^{\frac{3}{2}} x \sqrt{\cos \theta-\cot x \sin \theta}} dx=$$

$$\int \frac{\sec^2 x}{\sqrt{\tan x \cos \theta-\sin \theta}} dx+\int \frac{\operatorname{cosec}^2 x}{\sqrt{\cos \theta-\cot x \sin \theta}} dx$$

$$I=I_1+I_2 \quad ..... \text{ Let }\}$$

For I_1, let $\tan \mathrm{x} \cos \theta-\sin \theta=\mathrm{t}^2$

$\sec^2 x d x=\frac{2 t d t}{\cos \theta}$

For I_2, let $\cos \theta-\cot x \sin \theta=z^2$

$\operatorname{cosec}^2 x d x=\frac{2 z d z}{\sin \theta}$

$$I=I_1+I_2$$

$$=\int \frac{2 t d t}{\cos \theta t}+\int \frac{2 z d z}{\sin \theta z}$$

$$=\frac{2 t}{\cos \theta}+\frac{2 z}{\sin \theta}$$

$$=2 \sec \theta \sqrt{\tan x \cos \theta-\sin \theta}+2 \operatorname{cosec} \theta \sqrt{\cos \theta-\cot x \sin \theta}$$

Comparing

$AB=8 \operatorname{cosec} 2 \theta$